

Bakás

Leaving digital traces to navigate urban road hazards.

A lightweight, mobile-first Progressive Web App (PWA) that allows commuters, motorists, cyclists, and pedestrians to anonymously report and view urban road hazards in real-time. By utilizing crowdsourced civic traces, users map dangerous road traps such as potholes, open manholes, clogged drainages, and unlit streets with zero login barrier, resilient offline-first synchronization, and 5km PostGIS spatial filtering.

Target user	Main action	Stack	Design
Commuters, motorists, cyclists, and urban pedestrians	Zero-login 1-tap hazard reporting & 5km radar map inspection	React (TypeScript, Vite) / Leaflet.js / PostGIS / Supabase	Tactical Dark Slate Monochrome (#020617) Command Center

SECTION 00

Executive brief

A quality-first plan for a civic road hazard radar, not a bloated navigation app.

PRODUCT IN ONE SENTENCE

Bakás turns crowdsourced civic awareness into real-time, offline-resilient road hazard intelligence to protect urban commuters.

DECISIONS LOCKED FOR MVP

- **Frictionless Crowdsourcing (Zero-Login):** 1-tap anonymous reporting using browser GPS with under 5-second submission time.
- **Offline-First Resilience:** Local hazard reports saved in IndexedDB during cellular dead zones; synced automatically on reconnect.
- **Spatial Proximity Filter:** Strict 5-kilometer PostGIS radius filter ensures instant viewport loading and zero data bloat.
- **Automated Data Decay:** Time-decay TTL (24h temporary obstacles, 7d physical road defects) with community validation upvotes.
- **Dark Slate Monochrome System:** #020617 canvas with CartoDB Dark Matter tiles and severity radar pulses.

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SECTION 01

Product definition and real problem

Bakás is a real-time safety layer for the moment a commuter or motorist navigates urban streets.

THE SHARP PROBLEM STATEMENT

Urban commuters face dangerous, invisible road hazards (potholes, open manholes, flash floods, unlit streets) with no fast, lightweight way to report or avoid them in real-time.

Observed problem	Why it matters	Bakás response
Reporting apps require login, long forms, or deep menus.	Motorists and pedestrians abandon reporting while in transit.	Zero-login, 1-tap category + severity submission in under 5 seconds.
Cellular dead zones and weak signal in urban canyons cause data loss.	Hazards in dead zones (underpasses, rural corridors) go unrecorded.	IndexedDB offline-first queue with automatic background sync.
Outdated hazard pins remain indefinitely on civic maps.	Users lose trust in false alarms or resolved obstacles.	Automated TTL decay (24h/7d) extended only by live community validation.
Map rendering becomes slow and cluttered across entire cities.	High bandwidth consumption and slow mobile device rendering.	PostGIS 5km bounding circle strictly limits viewport rendering.

PRIMARY USER

A daily commuter, motorcyclist, delivery rider, cyclist, or pedestrian who needs immediate, glanceable road hazard awareness without friction.

POSITIONING

Bakás is	Bakás is not
A lightweight, anonymous urban road hazard radar	A turn-by-turn routing navigation app (like Waze/Google Maps)
An offline-first rapid civic reporting tool	An official government public works dispatch system
A glanceable dark slate command center	A social networking platform with profiles, feeds, or chat
A community-validated temporary trace network	A permanent static infrastructure database

SECTION 02

Users, needs, and product modes

Serve one clear core job while making room for different transit modalities.

User context	Need	Primary Bakás mode
"Napakadilim ng kalsada dito at may lubak."	Report an unlit road or deep pothole in 3 seconds before moving on.	Quick 1-Tap Trace Drop
"Bumabaha ba sa Dadaanang ko?"	Check for clogged drainage or flooded spots within 5km before leaving.	5km Proximity Radar View
"Nawalan ako ng signal sa ilalim ng tulay."	Record a road obstacle offline and have it sync automatically later.	Offline Dead-Zone Logging
"Naayos na ba yung bukas na manhole?"	Validate if a reported hazard is still present or mark it cleared.	Community Validation (Upvote / Clear)

CORE VALUE LOOP

Spotted Hazard → 1-Tap Anonymous Drop → Live 5km Radar Broadcast → Community Validation → Safer Commute.

PERSONALIZATION THAT BELONGS IN THE MVP

- **Proximity Radius Presets:** Instant toggle between 1km (hyperlocal walking), 3km (cycling), and 5km (motorcycle/driving).
- **Category Quick Filters:** 1-tap filtering for Potholes, Drainage/Flooding, Obstructions, and Dark Streets.
- **GPS Accuracy Ring:** Visual precision indicator showing device GPS accuracy radius.
- **Recentered Camera Lock:** Auto-tracking mode keeping user centered during transit.
- **Anti-Spam Fingerprint:** Client-side localStorage token preventing duplicate upvotes per device.

SECTION 03

End-to-end product workflow

A fast, frictionless sequence: locate, scan, pin, validate, and decay.

Stage	User action	Bakás response
01 / Locate	Opens PWA or desktop link.	Locks browser GPS in <1s and renders CartoDB Dark Matter map.
02 / Scan	Inspects 5km radar or applies filter.	Fetches and displays active hazards within 5km via PostGIS.
03 / Pin	Taps '+ Report Hazard' FAB button.	Opens tactile bottom sheet with 4 categories and severity picker.
04 / Submit	Selects category & severity, taps 'Submit'.	Optimistically drops pulsing pin; writes IndexedDB & syncs API.
05 / Validate	Nearby commuter taps pin on map.	Displays detail sheet; user can tap 'Still Here (+1)' or 'Cleared'.
06 / Decay	No action required (automated).	Expired pins automatically disappear from map after TTL ends.

DETAILED WORKFLOW CONTRACT

Input	Output	Exit gate
Device Geolocation coords	Validated lat/lng + accuracy radius	Coordinates within valid terrestrial bounds.
1-tap Category + Severity	Validated HazardPayload object	Non-null enum values with timestamp.
Local IndexedDB Write	Optimistic Radar DivIcon on map	UI updates in under 20ms without network wait.
Supabase PostGIS Sync	GEOGRAPHY Point (SRID 4326)	Record inserted with automated TTL calculation.

CORE UX RULE

Never block a user behind login screens, forms, or network timeouts when reporting a road hazard. Speed and safety are paramount.

SECTION 04

What Bakás produces

The value is real-time, actionable road intelligence, not static clutter.

HAZARD RECORD DATA CONTRACT

Field	Type	Purpose
id	UUID (v4)	Unique hazard identifier generated locally or by DB.
category	Enum	'pothole' 'clogged_drainage' 'road_obstruction' 'dark_street'
severity	Enum	'low' (caution) 'medium' (warning) 'high' (danger)
location	GEOGRAPHY(Point, 4326)	High-precision spatial coordinate point for PostGIS queries.
upvotes	Integer	Community validation counter extending pin TTL.
expires_at	Timestamp with TZ	Automated decay cutoff time (24h temporary, 7d physical).
sync_status	Local State	'synced' 'pending_sync' 'failed' (IndexedDB state).

RADAR RESULT EXPERIENCE

- **Glanceable Radar Markers:** Custom Leaflet DivIcons styled with SVG radar rings and pulsing animations for high severity.
- **Proximity Awareness:** Displays exact distance from user's current GPS location (e.g., '180m away').
- **Live TTL Countdown:** Dynamic time-to-decay display (e.g., 'Expires in 18 hrs' or 'Expires in 5 days').
- **Community Validation Feedback:** Real-time upvote count with optimistic instant incrementing.

RADAR HUD MODE

Full-screen 100dvh map with non-blocking floating controls, high-contrast dark theme (#020617), and screen wake-lock support for bike/car mounts.

SECTION 05

Spatial & offline architecture

Combine local IndexedDB storage, PostGIS spatial queries, and Service Worker caching.

Stage	Responsibility	Structured result
Geolocation Watcher	Tracks device position with high accuracy and error fallback.	UserLocation { lat, lng, accuracy }
IndexedDB Engine	Stores offline reports and cached 5km hazards locally in browser.	LocalHazardQueue[]
PostGIS Radius Query	Executes ST_DWithin spatial query on PostgreSQL via Supabase RPC.	ActiveHazardsWithin5km[]
Tile Caching Service Worker	Caches CartoDB Dark Matter map tiles for offline visual rendering.	CachedTileResponse
Sync Queue Manager	Listens for online event and flushes pending reports to database.	SyncResult { pushed, failed }

REQUEST SEQUENCE

1. App initializes → Service worker serves cached app shell & map assets.
2. Browser GPS acquires user location → queries PostGIS via `get\_hazards\_in\_radius(lat, lng, 5000)`.
3. When user reports hazard → written immediately to IndexedDB `pending\_reports` table.
4. If online → background fetch dispatches payload to Supabase; on success, mark as synced.
5. If offline → UI shows amber sync badge; Service worker retries upon reconnect.

ARCHITECTURAL BOUNDARY

Client handles fast UI rendering, GPS tracking, and IndexedDB persistence. Supabase handles PostGIS spatial indexing, anonymous RLS security policies, and automated TTL decay expiration.

SECTION 06

Supabase data and security plan

Strict spatial indexing and anonymous civic privacy without storing personal tracking data.

Entity	Important fields	Purpose
hazards	id, category, severity, location, upvotes, expires_at, created_at	Core PostGIS spatial table holding active road hazards.
hazard_validations	id, hazard_id, action_type ('upvote' 'resolve'), device_hash, created_at	Audit trail preventing duplicate upvotes per device.
spatial_index	GIST(location)	PostGIS GIST spatial index for sub-10ms 5km radius queries.

SECURITY & PRIVACY REQUIREMENTS

- **Zero-Login Row Level Security (RLS):** Anonymous public ``SELECT`` on non-expired hazards (``expires_at` > NOW()``).
- **Rate-Limited Anonymous Inserts:** Public ``INSERT`` allowed with spatial deduplication and rate-limiting triggers.
- **Zero User Tracking / PII:** Bakás never stores user identity, email, IP addresses, or historical GPS breadcrumbs.
- **Anti-Tampering Constraints:** Check constraints enforce valid enum values (``category``, ``severity``) and coordinate bounds.
- **Automated Decay Stored Procedure:** Scheduled cron job or RPC function to purge stale expired hazards automatically.

SECTION 07

MVP boundaries

One sharp civic workflow: locate, report a hazard in 5s, view 5km radar, and sync offline.

Build now	After the core flow	Explicitly out of scope
• Zero-login 1-tap hazard reporting	• Photo attachment for hazards	• Full turn-by-turn route navigation
• 5km PostGIS radius radar map	• Push notifications for nearby hazards	• User profiles, social feeds, and chats
• CartoDB Dark Matter Leaflet canvas	• Voice-activated hazard reporting	• Government agency dispatch CRM
• IndexedDB offline-first sync queue	• Heatmap density visualizations	• Commercial ad banners / monetization
• Community upvotes & auto-decay TTL	• PWA desktop dock integration	• Complex polygon hazard geofences

MVP ACCEPTANCE CHECKLIST

- First-time user can open the app and see active hazards within 5km in under 2 seconds.
- Reporting a hazard takes under 5 seconds with zero text input required.
- Offline report creation stores correctly in IndexedDB and syncs to Supabase upon reconnecting.
- Map panning triggers debounced PostGIS queries without UI stutter (60fps).
- Tapping 'Still Here (+1)' increments the counter and extends the pin's TTL.
- Mobile viewport layout operates cleanly with 1-handed thumb reachability.

SCOPE TEST

If a feature does not directly help a commuter report a road hazard in 5 seconds or view nearby hazards within 5km, it does not belong in the MVP.

SECTION 08

UI and visual direction

A tactical dark slate command center that delivers instant glanceability without night-blindness.

Element	Direction	Guardrail
Palette	Deep midnight slate (#020617), dark surface cards (#0f172a), high-contrast text (#f8fafc).	Never use saturated rainbow pins; use opacity and radar pulse frequency for severity.
Map Canvas	Leaflet.js styled with CartoDB Dark Matter tiles.	Tiles must blend seamlessly with background to make radar pins pop.
Typography	Modern geometric sans-serif (Inter) with crisp tracking for metrics.	Keep labels ultra-short and scannable.
Shape	Rounded-2xl bottom sheets, pill chips, tactical 1px slate-700 borders.	Avoid heavy shadows; rely on luminous borders and backdrop blur.
Motion	Radar pulse loops (1.8s) for high severity, smooth spring bottom sheets.	Respect prefers-reduced-motion with instant static states.

FIRST-VIEWPORT CONTENT

- **Top HUD:** Brand logo 'Bakás', Proximity Counter ('12 hazards in 5.0 km'), Sync Pill ('Online').
- **Radar Canvas:** Fullscreen interactive map with pulsing user GPS dot and nearby hazard pins.
- **Filter Chips:** Floating horizontal pills [All] [ Potholes] [ Drainage] [ Dark Streets].
- **Bottom Action HUD:** Recenter GPS Target + Elevated '+ Report Hazard' Action Pill.

CORE STATES

Empty: 'All clear within 5km radar.' | **Loading:** Radar sweep indicator | **Success:** Glowing hazard pins | **Offline:** Amber status badge with local queue | **Error:** GPS permission fallback guidance.

SECTION 09

Risks and mitigations

Anticipate spam, GPS inaccuracies, and offline edge cases before they affect users.

Risk or hard question	Honest answer	MVP mitigation
Will anonymous zero-login lead to spam pins?	Trolls or accidental clicks can create false pins.	15m deduplication rule, rate limits, and 3-vote community resolution decay.
Can urban GPS drift misplace pins?	Tall buildings and overpasses degrade GPS accuracy.	Display GPS accuracy circle and provide 1-tap drag-to-adjust pin mode.
Will offline sync create duplicate records?	Device might retry submissions multiple times.	Client-generated UUID v4 guarantees idempotency during batch sync.
What if obsolete pins stay on the map forever?	Resolved hazards ruin map credibility.	Strict automated TTL (24h/7d) removes inactive pins automatically.
Is reporting while driving dangerous?	Drivers looking at phones cause accidents.	1-tap big button UI; safety disclaimer advising passenger use only.

NON-NEGOTIABLE BOUNDARIES

- **Zero Account Friction:** Never introduce mandatory sign-up for viewing or reporting hazards.
- **Zero User Tracking:** Never log user movement routes or create location tracking histories.
- **Deterministic Decay:** All pins must have a finite `expires\_at` timestamp; no permanent pins.
- **Strict 5km Bound:** Queries must never fetch whole-country datasets into client memory.

SECTION 10

Validation and success criteria

Measure whether commuters navigate safely and report quickly, not vanity metrics.

Layer	Method	Minimum evidence
Problem Validation	Interview 5-8 daily motorcycle riders & commuters.	At least 4 report encountering unmapped potholes/floods weekly.
Workflow Speed	Time user from opening app to successful hazard drop.	Median report completion time is under 5 seconds.
Offline Resilience	Simulate airplane mode, drop 3 pins, restore network.	100% of offline reports sync to PostGIS without loss.
Spatial Accuracy	Test PostGIS radius query with 500 mock pins.	Query execution time <50ms with 0 false positives outside 5km.
Performance	Lighthouse mobile audit & 60fps pan benchmark.	Performance score >90; smooth 60fps tile and marker rendering.

CORE SUCCESS METRICS

- **Report Velocity:** Time to submit hazard trace < 5.0 seconds.
- **Offline Sync Rate:** 100% successful ingestion of queued IndexedDB reports upon reconnect.
- **Spatial Query Latency:** PostGIS 5km bounding box query < 100ms on 4G connections.
- **Decay Accuracy:** 0 stale pins remaining past their designated TTL window.

TEST INTEGRITY FIXTURES

Freeze golden test fixtures for 4 core urban scenarios: (1) EDSA deep pothole, (2) España Blvd clogged drainage flood, (3) Katipunan unlit street section, and (4) C-5 road debris obstruction.

SECTION 11

Build phase map: foundation

The first eight phases establish clarity, tactical visual identity, and a robust static frontend.

Phase	Outcome	Key deliverables	Exit gate
1. Product Definition	Stable civic promise & trust boundary.	Brief, non-goals, trust copy, golden fixtures.	MVP fits in one sentence.
2. Scope & Success	Realistic 5km offline-first MVP.	MVP contract, criteria, deferred features.	Every feature is now, next, or out.
3. Hazard Domain Research	Taxonomy of road hazards & decay rules.	Hazard categories, TTL matrix, risk log.	Each category has positive/negative rules.
4. Information Architecture	Complete product map & state matrix.	HUD shell map, bottom sheets, state inventory.	No undefined core state.
5. UX Wireframes	Understandable 5s flow before polish.	Wireframes, copy deck, thumb-zone layout.	User identifies FAB, report, and pins.
6. Tactical Visual System	Dark slate monochrome command center.	Tokens (#020617), CartoDB theme, radar pulses.	Contrast and night-vision pass.
7. Technical Foundation	Maintainable React + TypeScript + Vite base.	App shell, Leaflet setup, types, scripts.	Install, build, lint, typecheck pass.
8. Static Radar UI	Full journey works with controlled mock data.	Interactive map, mock pins, bottom sheets.	Demo works without backend.

WORKING RULE

Do not connect Supabase or complex PostGIS queries to a map UI that is still shifting at the component and state level. Static UI is the first proof that the product is glanceable and intuitive.

SECTION 12

Build phase map: product depth and release

Add spatial queries, offline sync, community validation, and PWA capabilities in careful sequence.

Phase	Outcome	Key deliverables	Exit gate
9. Hazard Data Contract	Stable schema for hazard records & TTL.	TypeScript schemas, validators, mock fixtures.	Fixtures render without runtime hacks.
10. PostGIS Viewport Engine	5km proximity spatial query handler.	Leaflet bounds listener, debounced fetcher.	Pan & zoom queries render smoothly.
11. Offline IndexedDB Sync	Zero data loss in cellular dead zones.	IndexedDB queue, sync manager, offline toast.	Offline reports sync on reconnect.
12. Supabase PostGIS API	PostgreSQL spatial backend & RLS.	Migrations, ST_DWithin RPC, anonymous RLS.	Spatial RPC responds in <100ms.
13. Validation & Decay	Community upvoting & automated TTL decay.	Upvote RPC, anti-spam hash, decay cron.	Upvotes extend TTL; expired pins decay.
14. PWA Native Hardening	Installable app with offline tile caching.	Manifest, Service Worker, GPS wake-lock.	PWA installs and opens offline.
15. Safety & Performance	Hardened, accessible, 60fps product.	WCAG 2.1 AA audit, rate limits, 60fps pass.	No critical safety/a11y defects.
16. Production Release	Live, documented, portfolio-ready release.	Vercel deployment, README, demo script.	Fresh reviewer tests in 5 mins.

DEPENDENCY RULE

Offline sync, PostGIS spatial queries, and community upvoting are valuable only after the basic map canvas, tactical dark UI, and local reporting flows are rock solid.

SECTION 13

Detailed phases 1-6

Define product boundaries, hazard taxonomy, and visual design before writing code.

PHASE 1: PRODUCT DEFINITION AND TRUST BOUNDARY

Outcome: One stable civic promise, target user profile, and honest zero-login boundary.

Workstreams: Define commuter & motorcyclist use cases; establish anonymous privacy rules; create golden fixtures.

Deliverables: Product brief, non-goals, trust copy, initial hazard fixture set.

Exit Gate: A first-time user understands Bakás in 30 seconds.

PHASE 2: SCOPE AND SUCCESS CRITERIA

Outcome: A tightly scoped, testable MVP contract.

Workstreams: Prioritize 5km radar, 1-tap reporting, IndexedDB offline sync; defer photo uploads and routing.

Deliverables: MVP feature contract, acceptance checklist, deferred feature roadmap.

Exit Gate: Every feature is classified as Now, Next, or Out.

PHASE 3: ROAD HAZARD DOMAIN RESEARCH

Outcome: Grounded taxonomy for urban road hazards, PostGIS coordinate systems, and decay logic.

Workstreams: Define 4 core categories (Pothole, Drainage, Obstruction, Dark Street); establish 24h/7d TTL rules.

Deliverables: Category schema, TTL decay matrix, spatial accuracy notes, safety risk log.

Exit Gate: Each hazard category has clear TTL, severity rules, and deduplication thresholds.

PHASE 4: INFORMATION ARCHITECTURE AND USER FLOW

Outcome: Complete application shell map from GPS lock to hazard submission.

Workstreams: Map HUD shell, Report bottom sheet, Hazard detail modal, Filter drawer, and Offline Sync manager.

Deliverables: Information architecture blueprint, screen inventory, complete state matrix.

Exit Gate: No screen depends on an undefined loading or error state.

PHASE 5: UX WIREFRAMES AND INTERACTION CONTRACT

Outcome: An intuitive 1-handed mobile flow designed for rapid thumb reachability.

Workstreams: Wireframe bottom action HUD, category 2x2 grid, severity selector, and validation buttons.

Deliverables: Low-fidelity wireframes, copy deck, touch target ergonomic notes.

Exit Gate: User can identify the Report button and submit a test pin in under 5 seconds.

PHASE 6: TACTICAL VISUAL DESIGN SYSTEM

Outcome: A distinct Dark Slate Monochrome command center aesthetic (#020617) with radar luminescence.

Workstreams: Finalize slate color tokens, CartoDB Dark Matter tile theme, radar pulse keyframes, and typography.

Deliverables: Tailwind design tokens, custom SVG marker DivIcons, typography scale, component specs.

Exit Gate: Contrast ratios pass WCAG AAA; radar pins are distinct without rainbow clutter.

SECTION 14

Detailed phases 7-11

Build the frontend base, static radar map, data schemas, PostGIS queries, and offline engine.

PHASE 7: TECHNICAL FOUNDATION AND TOOLING

Outcome: A clean, type-safe React 19 + TypeScript + Vite workspace.

Workstreams: Configure Vite, Tailwind CSS, Leaflet.js, Lucide icons; set up linting, formatting, and build scripts.

Deliverables: Clean repository scaffold, tsconfig, package scripts, Leaflet CSS integration.

Exit Gate: `npm run build` and `npm run lint` execute cleanly with zero errors.

PHASE 8: STATIC RADAR UI AND STATE COVERAGE

Outcome: Complete clickable map interface operating on mock fixture data.

Workstreams: Build `MapRadarCanvas`, `TopHUD`, `CategoryFilterBar`, `ReportBottomSheet`, `DetailBottomSheet`.

Deliverables: Interactive dark map, custom pulsing DivIcons, bottom sheet gestures, state mocks.

Exit Gate: Full user journey demos cleanly without external backend dependencies.

PHASE 9: HAZARD DATA CONTRACT AND FIXTURE ENGINE

Outcome: Strict TypeScript data schemas and adapters preventing malformed data.

Workstreams: Define `Hazard`, `HazardPayload`, and `ValidationAction` types; build TTL decay calculation helpers.

Deliverables: Data schema module, validation functions, golden fixtures for Manila urban road tests.

Exit Gate: Runtime validator safely rejects malformed coordinates or invalid enum values.

PHASE 10: POSTGIS SPATIAL VIEWPORT ENGINE

Outcome: Dynamic 5km proximity query engine with smooth Leaflet bounds handling.

Workstreams: Implement debounced map moveend listener; calculate bounding boxes; render clustered pins.

Deliverables: `useHazardViewport` hook, spatial distance calculators, marker cluster optimization.

Exit Gate: Panning and zooming triggers smooth fetches with zero frame drops (60fps).

PHASE 11: OFFLINE-FIRST ENGINE AND INDEXEDDB SYNC

Outcome: 100% offline resilience for reports created in cellular dead zones.

Workstreams: Integrate `idb` IndexedDB wrapper; build offline queue; implement auto-sync manager on `window.online`.

Deliverables: `offlineStorage` service, `useSyncManager` hook, offline status UI badge.

Exit Gate: Offline submissions persist in IndexedDB and flush to API upon reconnect.

SECTION 15

Detailed phases 12-16

Connect Supabase PostGIS, automate decay, harden PWA capabilities, and execute release verification.

PHASE 12: SUPABASE POSTGIS BACKEND AND SPATIAL API

Outcome: Production PostgreSQL database with PostGIS spatial indexing and anonymous RLS.

Workstreams: Write SQL migrations for `hazards` table (GEOGRAPHY Point); create `get\_hazards\_in\_radius` RPC; set RLS.

Deliverables: Database migration scripts, PostGIS RPC functions, Supabase client integration.

Exit Gate: Spatial radius RPC responds in under 50ms for 5km bounding queries.

PHASE 13: COMMUNITY VALIDATION AND DECAY AUTOMATION

Outcome: Live community upvoting ('Still Here') and automated TTL data decay.

Workstreams: Implement `upvote\_hazard` RPC; calculate TTL extension (+12h/+48h); set client fingerprinting to prevent vote spam.

Deliverables: Validation services, decay worker function, client anti-spam fingerprinting module.

Exit Gate: Upvoting extends `expires\_at`; 3 'Resolved' flags fade and decay the pin.

PHASE 14: PWA NATIVE HARDENING AND TILE CACHING

Outcome: Installable Progressive Web App with offline tile caching and wake-lock.

Workstreams: Configure Web App Manifest, Service Worker Workbox caching for CartoDB tiles, screen wake-lock API.

Deliverables: `manifest.webmanifest`, `sw.js` tile caching strategy, PWA install prompt banner.

Exit Gate: App installs cleanly on Android/iOS and opens reliably with zero internet connection.

PHASE 15: SAFETY, ACCESSIBILITY, AND PERFORMANCE REVIEW

Outcome: A resilient, accessible, and high-performance civic utility.

Workstreams: Run Lighthouse audit; verify outdoor sunlight contrast; test keyboard & screen reader navigation.

Deliverables: QA test report, accessibility remediation notes, rate-limiting security check.

Exit Gate: Lighthouse score >90; zero critical accessibility or safety blockers.

PHASE 16: PRODUCTION DEPLOYMENT AND PORTFOLIO PROOF

Outcome: Live Vercel deployment, comprehensive documentation, and structured demo flow.

Workstreams: Deploy to Vercel with HTTPS; configure Supabase environment secrets; write comprehensive README.

Deliverables: Live production URL, documentation suite, walkthrough guide, portfolio demo rehearsal.

Exit Gate: A fresh reviewer can test the live app and report a hazard in under 5 minutes.

SECTION 16

Demo story and decision log

The strongest portfolio story proves that Bakás delivers instant civic safety without friction.

Time	Action	What the audience learns
0:00 - 0:20	Open Bakás → instant GPS lock on dark radar map.	Zero-login entry; tactical dark command center aesthetic.
0:20 - 0:45	Inspect 5km proximity hazards and toggle category chips.	PostGIS 5km spatial filtering loads only relevant nearby hazards.
0:45 - 1:15	Tap '+ Report Hazard' → select 'Pothole' → tap 'Submit'.	1-tap reporting takes under 5 seconds with zero typing.
1:15 - 1:45	Toggle Airplane Mode → drop 'Clogged Drainage' pin.	IndexedDB saves report offline; UI shows amber sync queue badge.
1:45 - 2:15	Re-enable network → watch automatic background sync.	Offline report syncs to Supabase; pin updates to verified status.
2:15 - 2:40	Tap existing hazard → tap 'Still Here (+1)'.	Community validation extends TTL and increases marker pulse.
2:40 - 3:00	Showcase PostGIS architecture and automated data decay.	Product is built with extreme engineering rigor and civic safety focus.

DECISION LOG

Decision	Reason
React + TypeScript + Vite	Lightning-fast load times, strict type safety, and clean component modularity.
Leaflet + CartoDB Dark Matter	Lightweight tile rendering with perfect tactical dark aesthetic and low memory footprint.
Supabase PostgreSQL + PostGIS	Native spatial geometry types (`ST_DWithin`) enabling ultra-fast proximity queries.
IndexedDB Offline Queue	Guarantees zero data loss when users report hazards in cellular dead zones or tunnels.
Zero-Login Architecture	Eliminates all barrier to entry; road safety reporting must be instantaneous for everyone.
Automated Data Decay (24h/7d)	Prevents stale 'ghost hazards' and maintains high community trust in active map pins.

FINAL SCOPE TEST

If a feature does not directly empower an urban commuter to report a hazard in 5 seconds or view nearby hazards within 5km, it should not be part of the first Bakás release.